



**RAIN ENHANCEMENT
TECHNOLOGIES**

OPERATIONS REPORT: LA SAL, UT (WA25001)

Preliminary - February 2025

Version 1.0



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1 *Control Page*

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Revision History

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0.0c	Original Document Development – Draft	Jeffrey Chagnon; Rutuja Dongre; Scott Morris	02/12/2026
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2 *Executive Summary*

This report provides a preliminary review of the Weather Enhancement Technology Array (WETA) installed by Rain Enhancement Technologies (RET) on the Flat Iron Mesa, outside Moab, Utah. WETA is a self-sufficient and “off grid” solution installed to provide enhancement of precipitation in year-round operations (snow and rainfall) over the La Sal Mountain Ranges. The focus of this report is on the January 2026 operating period.

2.1 Operational Summary

1. This report covers 31 days from 01/01/2025 to 01/31/2025.
 - a. WETA operated on nine days in January 2026.
 - b. Precipitation was observed on seven of those days.
2. Like December 2025, January 2026 was a climatologically dry month.
3. Snow events on 1-2 January, 8-9 January, and 23-24 January replenished some of the snowpack that had degraded through December.
4. The evidence presented in this report supports the likelihood that precipitation was enhanced by WETA over the La Sal range.
5. Radiometrics radiometer data is being processed. Through consultation with Dr. Binod Pokharel, radiometer-derived quantities will be incorporated in forthcoming analyses as covariates.
6. Power issues related to the radiometer elaborated in the previous report were addressed in early February.
7. Addition of load shedding and backup generator for improved power reliability on site completed in early February.

2.2 Actions and Improvements

1. January precipitation, snow pack depth, and snow-water equivalent were evaluated against observations (both current and historical) from the nearby Abajo Mountains. Expectations for untreated snowpack were deduced through analysis snowpack in the La Sal Mountains compared to the Abajo in past years with similarly dry conditions.
2. Installation of further instrumentation in La Sal Ranges (planned Mar/Apr 2026).
3. Treatment probability maps are being calculated using HySPLIT simulations and estimated cell drift. A full event-based catalog of spatially-varying treatment probabilities will be presented in the next reporting cycle.
4. WETA operations will continue to target predicted precipitation events that are not considered adverse weather conditions or impacting snow pack.



3 *Operations Summary*

Figure 1 details the operating schedule during the reporting period. WETA was operated during periods when forecasts indicated any chance of precipitation over the target area, including marginal cases involving scattered light snow showers.

DATE(S)	WETA (ON/OFF)
01/01/2026	2157 off
01/02/2026 – 01/03/2026	off
01/04/2026	1939 on
01/05/2026	1134 off / 2534 on
01/06/2026	1159 off
01/07/2026	1605 on
01/08/2026	on
01/09/2026	1243 off
01/10/2026-01/22/2026	off
01/23/2026	0317 on
01/24/2026	1534 off
01/25/2026-01/31/2026	off

Figure 1: WETA operating schedule during January 2026. Green shading indicates periods of operation.

4 *Data and Analysis*

4.1 Summary of Precipitation Events

January 2026 was another relatively dry month. Figure 2 presents the daily time-series of precipitation measured at weather stations and SNOTEL in the region. Three main periods of snow occurred on 1-2 January, 8-9 January, and 23-24 January. WETA operated during these periods and had a likely impact on snowpack and snow water content.

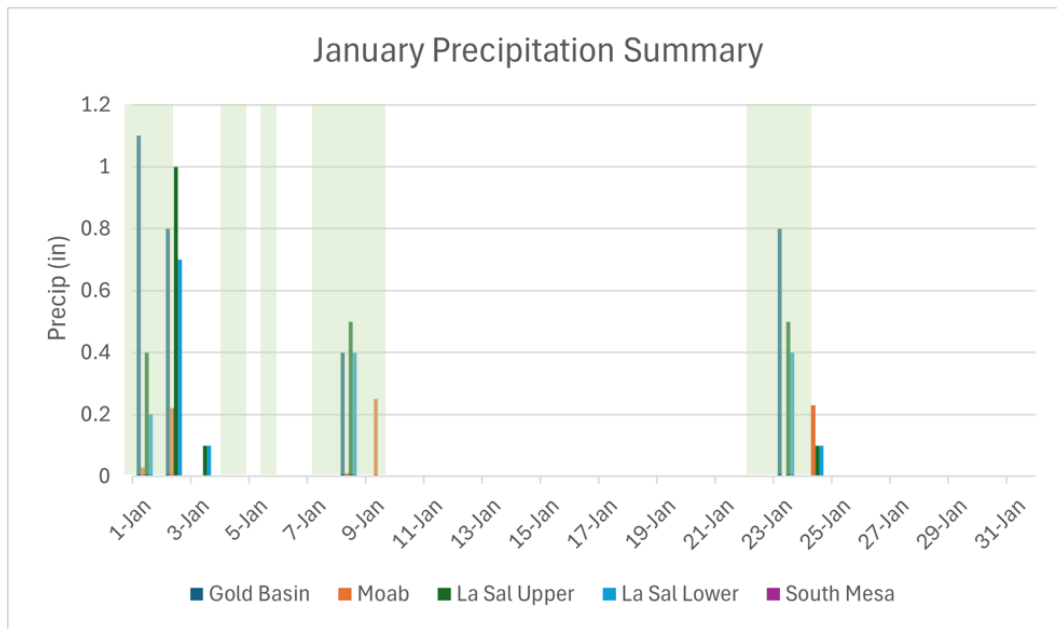


Figure 2: Summary of daily accumulated precipitation at reporting weather and SNOTEL stations.

4.2 Evaluation of Snow Pack and Snow-Water Content Enhancement

Snow pack enhancement was evident in the target area. The nearby Abajo mountains to the south of La Sal offer a means of establishing a control. We compared snow depth, snow-water equivalent, and accumulated precipitation at pairs of SNOTEL sites in the La Sal and Abajo ranges. Specifically, we compared observations from La Sal Upper to Buckboard Flat, and La Sal Lower to Camp Jackson. Figure 3 presents snow-water equivalent content at these pairs of sites, respectively. The comparison presented in Figure 3 reveals the following:

1. The climatology of snow water content at La Sal Upper is similar to that at Buckboard Flat. La Sal Lower maintains, on average, a shallower snowpack with less water content than Camp Jackson.
2. All sites had much less water in snowpack than average in December 2025 and January 2026 (see red circles in Fig.3).
3. Despite the dry conditions, the La Sal sites maintained a deeper snowpack with more water content than the Abajo sites. The differences were above the climatological mean.

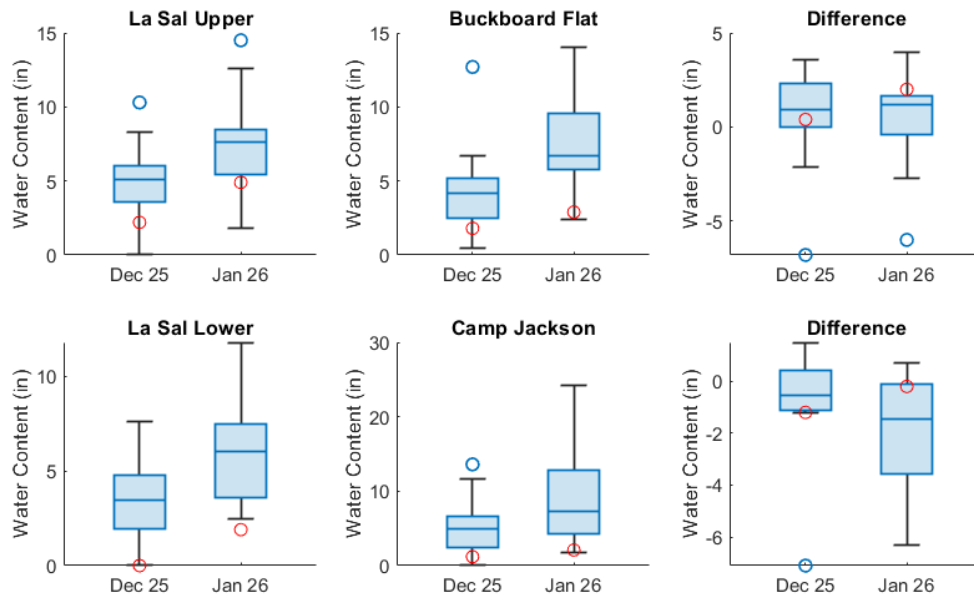


Figure 3: Box and whisker plots demonstrating SNOTEL-measured climatological December and January snow water content at (top left) La Sal Mountain Upper, (top middle) Buckboard Flat, (bottom left) La Sal Mountain Lower, and (bottom middle) Camp Jackson. The differences are shown in the right-most column. The red circles in each panel indicates values for the current year. Blue circles indicate climatological outliers.

4.3 Establishment of Controls: Comparison of January 2026 to Analog Years

While Figure 3 indicates that enhancement was likely, quantitative estimation of the enhancement requires controls that are appropriate for the historically dry conditions that were observed in January 2026. We therefore evaluated the January 2026 observations against analog years that were

exceptionally dry. Using precipitation accumulation as a basis for selection, the five driest years since 2013 were 2014, 2018, 2022, 2024, and 2025. The differences between observations in the La Sal and Abajo ranges during these years were used to estimate "expected differences". Differences observed in 2026 that exceeded those expected differences are a measure of "enhancement above expectation". Figure 4 summarizes the expected enhancement values for January 2026 using this methodology. At La Sal Upper, snow depth was likely enhanced by 3-4 inches and snow-water content was likely enhanced by approximately 2 inches at La Sal Upper. The comparison between La Sal Lower and Camp Jackson is subject to more variability, most likely due to the lower altitude and proximity to snowline.

	Snow Depth (in)	Snow-Water Eq (in)	Accum. Precip (in)
LA SAL UPPER	+3.6	+2	+0.74
LA SAL LOWER	+0.8	-0.94	+0.36

Figure 4: January 2026 enhancement estimated by taking the difference between observations at sites in La Sal and Abajo compared to the difference observed in the driest five recent years (2014, 2018, 2022, 2024, 2025).

4.4 Analysis of Radiometer Data

Radiometer-estimated liquid water content indicated significant differences in cloud liquid water over the La Sal range during the 23-24 January snow event. Fig. 5 presents the difference in vertically-integrated liquid water between two azimuthal directions – one to the northeast directed towards the La Sal range, and one directed vertically. Higher values over the La Sal range were evident during the snowfall period on the 23rd (e.g., see Fig. 2). RET has consulted with Dr Binod Pokharel on best use of the radiometer data. Based on that discussion, derived radiometer quantities will be used as covariates to evaluate whether LWC or supercooled LWC may moderate the efficacy of enhancement.

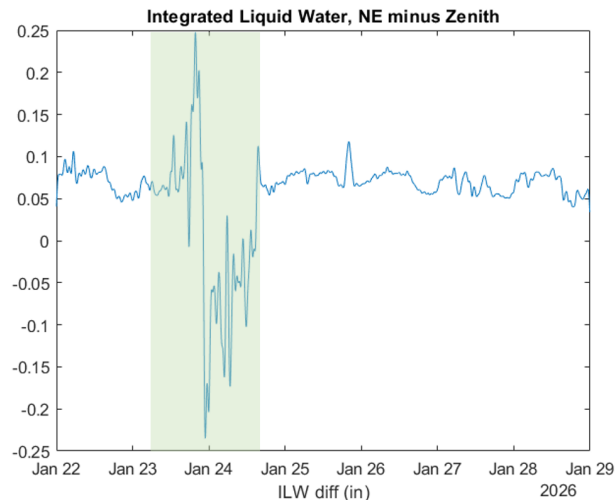


Figure 5: Low-pass filtered difference in radiometer-estimated vertically-integrated liquid water between two azimuthal directions: northeast (treatment) minus zenith (control).

5 *References*

5.1 Data Sources

Type of Data Source	Specific Locations	Data Description	Granularity/ Frequency
Radar	Grand Junction, CO	Radar reflectivity, velocity, precipitation accumulation, and all Level II and III fields at all elevation angles	~5 minutes
Radiometer	Moab, UT	Relative humidity, liquid water content in all scan directions	One minute
Weather Station	South Mesa, UT; Gold Basin, UT; Moab, UT; US-191 at MP 104 Flat Iron, UT; Hole N The Rock, UT; La Sal, UT; SR-46 at MP 12.5 La Sal Divide (UT DOT), UT	Weather variables such as precipitation, wind speed, wind direction, temperature, etc.	Hourly
SNOTEL	Gold Basin, UT (1304); La Sal Mtn, UT (572); La Sal Mtn Lower, UT (1215); Buckboard Flat, UT (1153); Camp Jackson, UT (383)	Snow / water equivalent monitoring	Daily